

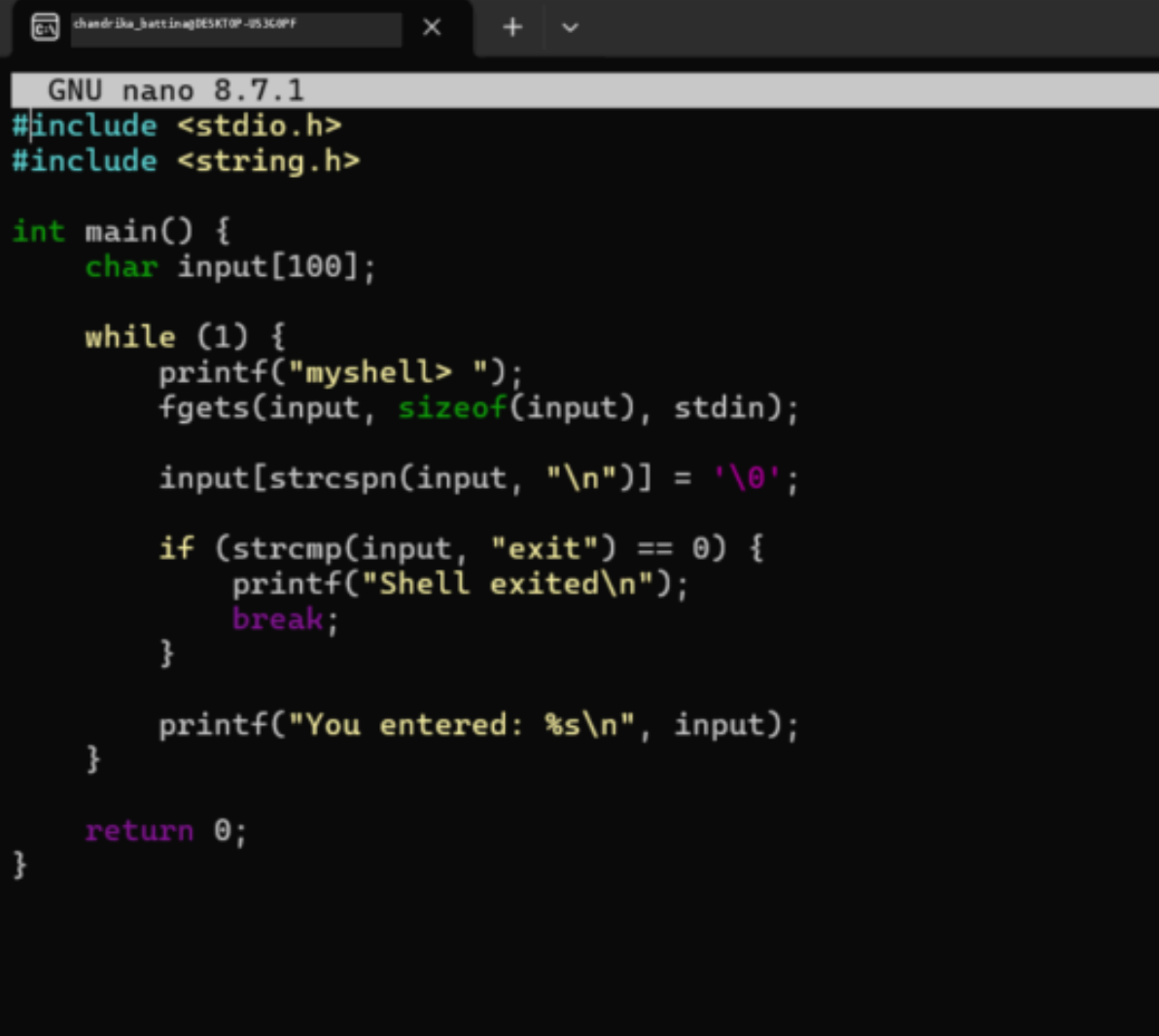
OSSP (Operating Systems And Systems Programming) - 25CS2104E

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1. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

1(a) Mainloop code:

A screenshot of a terminal window with a dark background. The window title bar shows 'chandr-bha_hettinagDESKTOP -WS3C/0FF'. The terminal displays the GNU nano 8.7.1 editor interface. The code is written in C and implements a simple shell mainloop. It includes headers for stdio and string, defines a main function, and uses a while loop to repeatedly prompt the user, read input, and process it. The code uses printf for prompts and output, fgets for reading input, and strcmp for checking the 'exit' command. The input buffer is managed by replacing the newline character with a null terminator. The code is as follows:

```
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    while (1) {
        printf("myshell> ");
        fgets(input, sizeof(input), stdin);

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0) {
            printf("Shell exited\n");
            break;
        }

        printf("You entered: %s\n", input);
    }

    return 0;
}
```

OUTPUT SCREENSHOT:

```
root@unknown:~# nano mainloop.c
root@unknown:~# gcc mainloop.c -o shell
root@unknown:~# ./shell
myshell>
You entered: hello Im sabss
myshell> ls
You entered: ls
myshell> exit
Shell exited
chandraka_battina@DESKTOP-US360PF1-8 ~$
root@unknown:~#
```

1(b) Capture Keyboard Input code:

```
#include <stdio.h>
#include <termios.h>
#include <unistd.h>

int main() {
    struct termios old, new;
    char ch;
    char buffer[100];
    int i = 0;

    tcgetattr(STDIN_FILENO, &old);
    new = old;
    new.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &new);

    printf("Enter command: ");

    while (1) {
        ch = getchar();

        if (ch == '\n') {
            buffer[i] = '\0';
            break;
        }

        if (ch == 127) {
            if (i > 0) {
                i--;
                printf("\b  ");
            }
        }

        else if (i < 99) {
            buffer[i++] = ch;
            printf("%c", ch);
        }
    }

    tcsetattr(STDIN_FILENO, TCSANOW, &old);

    printf("Command: %s\n", buffer);

    return 0;
}
```

OUTPUT SCREENSHOT:

```
root@unknown:~# nano keyboard.c
root@unknown:~# gcc keyboard.c -o shell
root@unknown:~# ./shell
Enter command: hellooo
Command: hellooo
root@unknown:~# gcc keyboard.c -o shell
root@unknown:~# ./shell
Enter command: hello
Command: hello
```